REST

Based on the codebase created during the previous module, implement follow REST API (as a RestController):

1. Trainee Registration (POST method)
    a. Request
        I. First Name (required)
        II. Last Name (required)
        III. Date of Birth (optional)
        IV. Address (optional)
    b. Response
        I. Username
        II. Password
2. Trainer Registration (POST method)
    a. Request
        I. First Name (required)
        II. Last Name (required)
        III. Specialization (required) (Training type reference)
    b. Response
        I. Username
        II. Password
3. Login (GET method)
    a. Request
        I. Username (required)
        II. Password (required)
    b. Response
        I. 200 OK
4. Change Login (PUT method)
    a. Request
        I. Username (required)
        II. Old Password (required)
        III. New Password (required)
    b. Response
        I. 200 OK
5. Get Trainee Profile (GET method)
    a. Request
        I. Username (required)
    b. Response
        I. First Name
        II. Last Name
        III. Date of Birth
        IV. Address
        V. Is Active
        VI. Trainers List

1. Trainer Username
2. Trainer First Name
3. Trainer Last Name
4. Trainer Specialization (Training type reference)

6. Update Trainee Profile (PUT method)
    a. Request
        I. Username (required)
        II. First Name (required)
        III. Last Name (required)
        IV. Date of Birth (optional)
        V. Address (optional)
        VI. Is Active (required)
    b. Response
        I. Username
        II. First Name
        III. Last Name
        IV. Date of Birth
        V. Address
        VI. Is Active
        VII. Trainers List
            1. Trainer Username
            2. Trainer First Name
            3. Trainer Last Name
            4. Trainer Specialization (Training type reference)
7. Delete Trainee Profile (DELETE method)
    a. Request
        I. Username (required)
    b. Response
        I. 200 OK
8. Get Trainer Profile (GET method)
    a. Request
        I. Username (required)
    b. Response
        I. First Name
        II. Last Name
        III. Specialization (Training type reference)
        IV. Is Active
        V. Trainees List
            1. Trainee Username
            2. Trainee First Name
            3. Trainee Last Name
9. Update Trainer Profile (PUT method)
    a. Request
        I. Username (required)
        II. First Name (required)
        III. Last Name (required)

IV. Specialization (read only) (Training type reference)
V. Is Active (required)

b. Response
I. Username
II. First Name
III. Last Name
IV. Specialization (Training type reference)
V. Is Active
VI. Trainees List
1. Trainee Username
2. Trainee First Name
3. Trainee Last Name

10. Get not assigned on trainee active trainers. (GET method)
a. Request
I. Username (required)
b. Response
I. Trainer Username
II. Trainer First Name
III. Trainer Last Name
IV. Trainer Specialization (Training type reference)

11. Update Trainee's Trainer List (PUT method)
a. Request
I. Trainee Username
II. Trainers List (required)
1. Trainer Username (required)
b. Response
I. Trainers List
1. Trainer Username
2. Trainer First Name
3. Trainer Last Name
4. Trainer Specialization (Training type reference)

12. Get Trainee Trainings List (GET method)
a. Request
I. Username (required)
II. Period From (optional)
III. Period To (optional)
IV. Trainer Name (optional)
V. Training Type (optional)
b. Response
I. Training Name
II. Training Date
III. Training Type
IV. Training Duration
V. Trainer Name

13. Get Trainer Trainings List (GET method)
a. Request

I. Username (required)
II. Period From (optional)
III. Period To (optional)
IV. Trainee Name (optional)

b. Response
I. Training Name
II. Training Date
III. Training Type
IV. Training Duration
V. Trainee Name

14. Add Training (POST method)
    a. Request
        I. Trainee username (required)
        II. Trainer username (required)
        III. Training Name (required)
        IV. Training Date (required)
        V. Training Duration (required)
    b. Response
        I. 200 OK
15. Activate/De-Activate Trainee (PATCH method)
    a. Request
        I. Username (required)
        II. Is Active (required)
    b. Response
        I. 200 OK
16. Activate/De-Activate Trainer (PATCH method)
    a. Request
        I. Username (required)
        II. Is Active (required)
    b. Response
        I. 200 OK
17. Get Training types (GET method)
    a. Request -no data
    b. Response
        I. Training types
            1. Training type
            2. Training type Id

Notes:

1. During Create Trainer/Trainee profile username and password should be generated as described in previous modules.
2. Not possible to register as a trainer and trainee both.

3. All functions except Create Trainer/Trainee profile. Should be executed only after Trainee/Trainer authentication (on this step should be checked username and password matching).
4. Implement required validation for each endpoint.
5. Users Table has parent-child (one to one) relation with Trainer and Trainee tables.
6. Training functionality does not include delete/update possibility via REST
7. Username cannot be changed.
8. Trainees and Trainers have many to many relations.
9. Activate/De-activate Trainee/Trainer profile not idempotent action.
10. Delete Trainee profile is hard deleting action and bring the cascade deletion of relevant trainings.
11. Training duration have a number type.
12. Training Date, Trainee Date of Birth have Date type.
13. Is Active field in Trainee/Trainer profile has Boolean type.
14. Training Types table include constant list of values and could not be updated from the application.
15. Implement error handling for all endpoints.
16. Cover code with unit tests.
17. Two levels of logging should be implemented:
    1. Transaction level (generate transactionId by which you can track all operations for this transaction the same transactionId can later be passed to downstream services)
    2. Specific rest call details (which endpoint was called, which request came and the service response - 200 or error and response message)
18. Implement error handling.
19. Document methods in RestController file(s) using Swagger annotations.